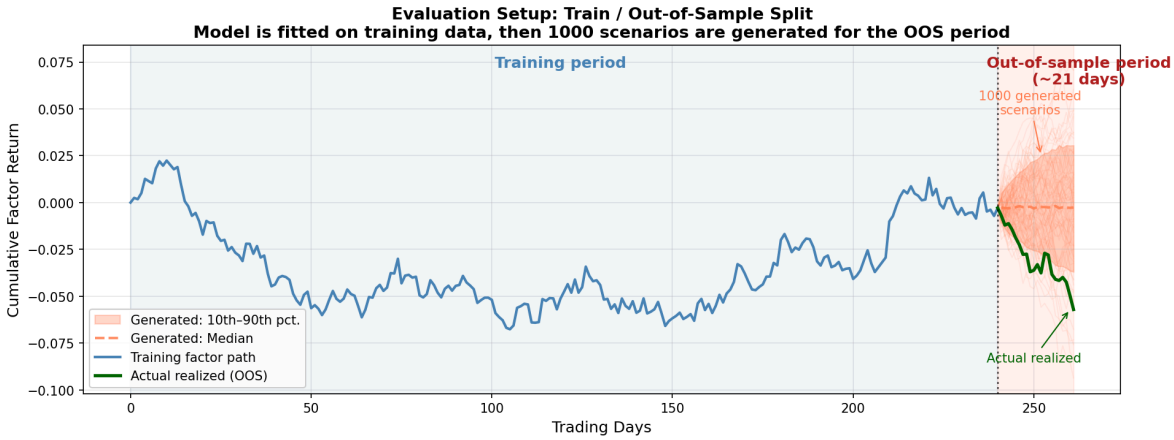

Visualization Guide for Model Evaluation

1. Evaluation Setup

The evaluation is built around a train/out-of-sample (OOS) split. The model is fitted on the training period only. We then use the fitted model to generate 1000 scenarios for the OOS period, which in our setting is approximately one month (about 21 trading days). At the end of that period we also have the actual realized returns, which we can compare against.

This gives us three sets of data to work with:

- **Training data**: the historical factor and stock returns used to fit the model. This is the distribution the model has "seen".
- **Generated scenarios**: 1000 draws produced by the model for the OOS period. Each draw is a plausible path according to what the model has learned.
- **Actual OOS**: the single realized path of factor and stock returns over the same OOS period.

This setup is illustrated below. The diagram shows one factor: the blue line is the cumulative training history, the fan of coral lines is the 1000 generated scenarios fanning out from the split date, the shaded band captures the 10th-90th percentile range of those scenarios, and the dark green line is what actually happened in the OOS period.

The key comparisons we care about are:

- **Generated vs training distribution**: does the model reproduce the distribution it was trained on? If not, something is wrong with the model or training procedure.
- **Generated vs OOS distribution**: does the model generalize to unseen data? Some divergence is expected and normal, because markets are non-stationary. But large, systematic differences are worth investigating.
- **Training vs OOS distribution**: are the two windows even comparable? If the market regime shifted between windows (e.g. the OOS period is a crisis and the training period was calm), you should expect differences in the data themselves, not just in the generated output.

---

2. Factor-Level Evaluation

Start here, because factors are lower-dimensional (40 factors vs 500 stocks) and easier to inspect individually. For each factor you have the training distribution, the OOS distribution, and 1000 generated draws. The generated draws for a single factor can be pooled across all 1000 scenarios to form one large sample.

2.1 Moment Comparison

Compute the mean, standard deviation, skewness, and kurtosis for each of the three sets (generated, training, OOS) for every factor. Present these as a scatter plot: one point per factor, with the actual moment on the x-axis and the generated moment on the y-axis. Do this separately for each moment. Points lying on the 45-degree diagonal indicate good reproduction.

- **Mean**: small non-zero means are common in financial data. Check whether the model is biased.
- **Standard deviation**: the most important moment. If the model is over- or under-dispersed this shows up immediately.
- **Skewness**: many factors have mild negative skew (losses are larger than gains). The model should capture this.
- **Kurtosis**: fat tails. If generated kurtosis is too low, the model is not producing enough extreme events.

2.2 Distribution Comparison (Histogram / KDE)

For a handful of representative factors (e.g. the market factor, a style factor, a currency factor), overlay:

- The histogram or KDE of the training distribution.
- The histogram or KDE of the OOS distribution.
- The histogram or KDE of the generated distribution.

This gives a more complete picture than moments alone. Pay particular attention to the tails.

2.3 Quantile-Quantile (Q-Q) Plots

Q-Q plots put the quantiles of two distributions against each other. If they match, the points lie on the diagonal. Deviations in the tails are easy to spot. Make a Q-Q plot of generated vs training for each factor (or at least a selection). If the tails curve away from the diagonal, the model is producing too light or too heavy tails relative to what it was trained on.

2.4 Kolmogorov-Smirnov Test

The KS test produces a p-value for the null hypothesis that two samples come from the same distribution. Rather than interpreting each factor's test individually, collect all 40 p-values and look at the distribution of p-values. Under the null (model reproduces the distribution perfectly), p-values should be roughly uniform between 0 and 1. If many p-values pile up near zero, the model is systematically off. Do this twice: once comparing generated vs training, and once comparing generated vs OOS.

2.5 Autocorrelation Structure

Factor returns often have mild autocorrelation (e.g. short-term momentum) or autocorrelation in squared returns (volatility clustering). Plot the autocorrelation function (ACF) for a few factors: training, OOS, and one sample generated scenario. If the model does not reproduce the temporal structure, the generated paths will look too noisy or too smooth compared to the actual paths.

---

3. Stock-Level Evaluation

With 500 stocks you cannot inspect each one individually. The goal is to summarize the quality of the stock-level generation across the entire cross-section.

3.1 Moment Scatter Plots

Compute the time-series moments (mean, std, skewness, kurtosis) for each stock, separately for the generated data (pooled across scenarios), the training data, and the OOS data. Then make one scatter plot per moment: one point per stock, actual on x-axis, generated on y-axis.

- A cloud of points around the diagonal means the model is reproducing stock-level properties well.
- A cloud that is systematically above or below the diagonal indicates a bias.
- For standard deviation especially: if the points lie on the diagonal, individual stock volatilities are being reproduced correctly. This is the time-series validation.

3.2 Distribution of KS Test P-values

Run a KS test for each stock (generated vs training, generated vs OOS), then plot the histogram of the resulting 500 p-values. The same logic as for factors applies: under the null, the p-values should be uniform. A spike at low p-values shows that for many stocks the distributions differ systematically.

---

4. Cross-Sectional Evaluation

The previous sections look at the time-series of each factor or stock. Now we want to look at the cross-section: at each point in time, what does the distribution across all stocks look like?

4.1 Cross-Sectional Dispersion

For each time point in the OOS period, compute the cross-sectional standard deviation of returns across all 500 stocks. Do this for the actual data (one number per date), and for each generated scenario (also one number per date per scenario). The result is a distribution of cross-sectional standard deviations for the generated scenarios, and a single sequence for the actual data.

Plot both as distributions (histogram or KDE) and compare. In particular, check the ratio of the mean generated cross-sectional std to the mean actual cross-sectional std. If this ratio is above 1, the model is generating returns that are too spread out across stocks.

Important: note that cross-sectional dispersion is different from time-series volatility. Even if individual stock volatilities are perfectly reproduced (which you can verify with the moment scatter plots above), cross-sectional dispersion can still be too wide. This happens because cross-sectional dispersion depends on how correlated the stocks are within a given day, not just on how volatile each stock is over time.

4.2 Decomposition into Systematic and Idiosyncratic Components

Because we use a factor model, every stock return can be decomposed into:

- **Systematic component**: the part explained by factor returns and loadings
- **Idiosyncratic component**: the residual unexplained by factors

You can compute this decomposition for both the actual returns and the generated returns. Then compare the cross-sectional distribution of actual vs generated at three levels: total return, systematic component, and idiosyncratic component.

This helps diagnose the source of any dispersion mismatch. If the systematic component is too wide, the factor returns themselves are too volatile or the loadings are too large. If the idiosyncratic component is too wide, the residual variance is too large. Knowing which component is the problem points to what needs to be fixed.

Plot:

- The distribution of cross-sectional std for each component (total, systematic, idiosyncratic) — generated vs actual side by side.
- Sample cross-sections: overlay a handful of generated cross-sections on top of actual ones to compare the shape visually.

---

5. Critical Thinking: Interpreting What You See

When you look at the results, do not just report whether numbers match or not. Think about why you see what you see.

Generated matches training but not OOS
This is the most common scenario and does not necessarily mean the model is bad. The training and OOS periods may simply have different properties. First check: are the training and OOS actual distributions themselves different? If yes, the model can only be expected to reproduce what it was trained on. The useful question then becomes: how large is the distribution shift, and is the model's coverage interval wide enough to include the OOS outcomes?

Generated does not match training
Something is wrong with the model: underfitting, insufficient training, wrong architecture, or unstable training. Check loss curves. Cross-check whether the training data itself was processed correctly.

Cross-sectional dispersion is too wide
This is an issue that was found in the parametric model as well (the generated cross-sectional standard deviation was consistently ~1.5x too wide). The model produces individual stock volatilities that look correct, but the cross-section is overdispersed. Decompose into systematic and idiosyncratic to find the source. Think about whether the factor loadings or residual variances are estimated on a different sample/frequency than the one being simulated.

Tails are too thin
If generated kurtosis is too low and the Q-Q plot curves upward at the extremes, the model is underrepresenting tail events. This is the main limitation of models with Gaussian assumptions. Check whether the training data itself contains enough extreme events, and whether the model has any mechanism (e.g. t-distribution residuals, volatility clustering) to generate tails.

Factor autocorrelation differs
If the generated factor ACF does not match the training data, the model may be treating each time step as independent. For the parametric model this is the expected behavior (factor draws are i.i.d.). For the diffusion model, temporal structure should be built in via conditioning.

---

6. Reference: MacroVAE and Other Papers

The MacroVAE paper evaluates generated scenarios in ways that go beyond comparing univariate distributions. When reading it, look specifically for:

- How they evaluate joint distributions (not just marginals for each factor/stock)
- How they incorporate downstream tasks into evaluation (e.g. portfolio construction, strategy performance on generated vs actual data)
- Any stylized-fact specific tests they propose: volatility clustering, leverage effect, fat tails

Other metrics commonly used in the generative finance literature:

- **Wasserstein distance / Earth Mover's Distance**: measures how much "work" it takes to reshape one distribution into another. More sensitive to the shape of the full distribution than the KS statistic.
- **Maximum Mean Discrepancy (MMD)**: compares distributions in a kernel-embedded space, works well for high-dimensional joint distributions.
- **Stylized fact tests**: ACF of returns (should be near zero), ACF of absolute or squared returns (should be positive and slowly decaying), correlation asymmetry, leverage effect.
- **Portfolio-level evaluation**: construct a portfolio on generated data and simulate its performance on actual data. Compare to the same portfolio built on historical data. This is an end-to-end check on whether the generated scenarios are useful for the actual downstream task.

Final note: visualization is iterative. Start with the standard plots listed here, but treat anything unexpected as a question rather than just a result. When something looks wrong, make a new plot specifically to diagnose the cause before drawing conclusions.